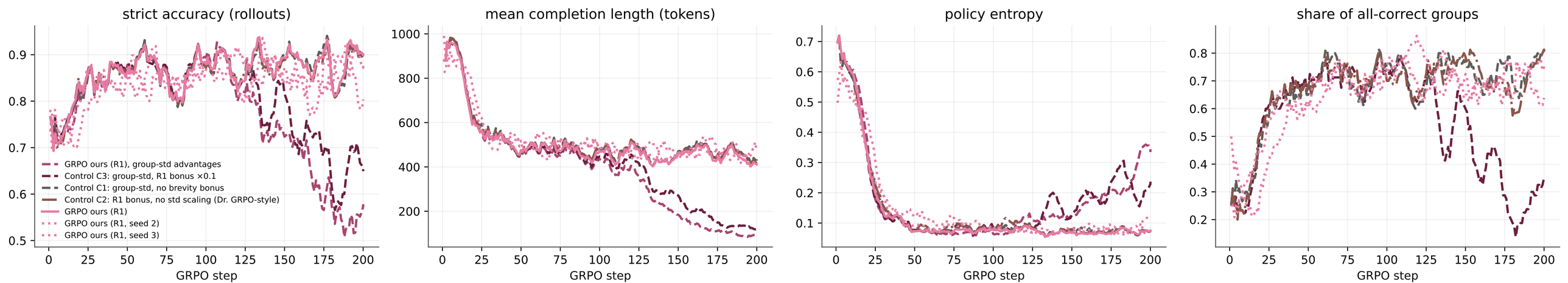
strict accuracy (rollouts)
mean completion length (tokens)
policy entropy
share of all-correct groups
GRPO step
GRPO step
GRPO step
GRPO step
GRPO ours (R1), group-std advantages
Control C3: group-std, R1 bonus ×0.1
Control C1: group-std, no brevity bonus
Control C2: R1 bonus, no std scaling (Dr. GRPO-style)
GRPO ours (R1)
GRPO ours (R1, seed 2)
GRPO ours (R1, seed 3)